

FM405 Fixed Income Securities and Credit Markets

Summative Assessment: Monte Carlo Mortgage Valuation

University Building Society (UBS) Mortgage Pricing Model

Candidate Numbers: 67125, 70216

Parameters: $T = 10$ years | $N = 100,000$ simulations | Semi-annual steps ($\Delta = 0.5$) | Principal = 100,000

Data source: Bank of England, UK Gilt Zero-Coupon Yield Curve, 30 January 2026

Models: Ho–Lee (level volatility $\sigma = 1.85\%$) | Black–Derman–Toy (log-rate volatility $\sigma = 20.00\%$)

1 Part (a): Construction and Calibration of the Interest Rate Trees

1.1 Objective and Framework

This section constructs arbitrage-free binomial trees for the risk-neutral short rate using the Ho–Lee and Black–Derman–Toy (BDT) models, calibrated to the observed UK zero-coupon gilt yield curve. Both models are implemented on a semi-annual grid ($\Delta = 0.5$) over a 10-year horizon (20 steps), matching the frequency of mortgage cash flows analysed in subsequent parts.

1.2 Data Source and Yield Curve

The term structure is taken from the Bank of England zero-coupon gilt curve on 30 January 2026. Gilt yields provide a market-based benchmark for the risk-free term structure in the UK. Using zero-coupon yields is convenient because they map directly into discount factors without bootstrapping. The observed curve is upward sloping, with the continuously compounded zero rate increasing from 3.48% at 0.5 years to 4.63% at 10 years, consistent with expectations of rising future short rates and/or a positive term premium. Table 1 reports the full yield curve and corresponding zero-coupon bond (ZCB) prices.

Table 1: Input Yield Curve (Bank of England, 30 January 2026)

Maturity	Yield (%)	ZCB Price
0.5	3.4800	98.2751
1.0	3.5500	96.5123
1.5	3.5900	94.7574
2.0	3.6300	92.9973
2.5	3.6700	91.2333
3.0	3.7200	89.4402
3.5	3.7800	87.6078
4.0	3.8400	85.7615
4.5	3.9000	83.9037
5.0	3.9700	81.9960
5.5	4.0400	80.0755
6.0	4.1100	78.1453
6.5	4.1800	76.2083
7.0	4.2500	74.2673
7.5	4.3100	72.3793
8.0	4.3800	70.4406
8.5	4.4500	68.5059
9.0	4.5100	66.6377
9.5	4.5700	64.7815
10.0	4.6300	62.9393

All rates are continuously compounded throughout. The initial short rate is set to $r_0 = 3.48\%$, i.e. the 6-month zero-coupon yield, which is the natural proxy for the current short rate in a semi-annual discretisation.

1.3 Volatility Estimation

The yield curve determines discount factors today but not the future distribution of short rates, which requires a volatility assumption. For the Ho–Lee model, the volatility parameter $\sigma_{\text{HL}} = 1.85\%$ represents the annualised standard deviation of changes in the level of the short rate. This is estimated from monthly changes in the Bank of England Bank Rate over the period January 2015 to December 2025 (121 observations), yielding a monthly standard deviation of approximately 0.53% , which annualises to 1.85% . This is close to the lecture benchmark of 1.73% (US data, Veronesi Table 11.1) and is appropriate for the recent UK rate environment.

For the BDT model, $\sigma_{\text{BDT}} = 20.00\%$ represents the annualised volatility of log short rates. This is estimated from monthly log-differences in the Bank Rate over the same sample, consistent with the lecture methodology (Veronesi, Chapter 11), where an analogous US estimation yields $\sigma = 21.42\%$. The slightly lower value reflects the shorter and more

stable UK sample period.

1.4 Calibration Procedure

Calibration follows the recursive procedure from Chapter 3 of the lecture notes (Kondor, Chapter III, slides 6–9 for Ho–Lee; slides 21–22 for BDT; see also Veronesi (2010), Section 11.2). At each time step i , a drift parameter θ_i is chosen so that the tree exactly reprices the zero-coupon bond maturing at step $i + 2$. Specifically, θ_i is found by numerical root-finding (Brent’s method) such that the backward-induction price of the $(i + 2)$ -step ZCB equals its observed market price. The risk-neutral probability is fixed at $q = \frac{1}{2}$ throughout. This approach mirrors the lecture example on January 8, 2002 data, extended here from $T = 5$ to $T = 10$ years.

For Ho–Lee (Chapter III, slide 6), the short rate evolves as:

$$r_{i+1,j} = r_{i,j} + \theta_i \Delta + \sigma \sqrt{\Delta} \quad (\text{up}), \quad (1)$$

$$r_{i+1,j+1} = r_{i,j} + \theta_i \Delta - \sigma \sqrt{\Delta} \quad (\text{down}). \quad (2)$$

Shocks are additive in rate levels.

For BDT (Chapter III, slide 21), the same structure applies to the log short rate $z_{i,j} = \ln(r_{i,j})$:

$$z_{i+1,j} = z_{i,j} + \theta_i \Delta + \sigma \sqrt{\Delta} \quad (\text{up}), \quad (3)$$

$$z_{i+1,j+1} = z_{i,j} + \theta_i \Delta - \sigma \sqrt{\Delta} \quad (\text{down}), \quad (4)$$

with $r_{i,j} = \exp(z_{i,j})$, guaranteeing positive rates. As noted on slide 22, σ for BDT must be estimated from a log interest rate series.

Table 2: Calibrated Drift Parameters θ_i ($\times 100$)

Step i	0	1	2	3	4	5	6	7	8	9
t	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5
Ho–Lee	0.2886	0.1257	0.2028	0.2199	0.3570	0.4341	0.3512	0.3682	0.5853	0.4424
BDT	5.9307	0.8591	2.5059	2.4975	5.5463	6.8535	4.2899	4.2429	8.6164	4.9487

Step i	10	11	12	13	14	15	16	17	18
t	5.0	5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0
Ho–Lee	0.4594	0.4764	0.4934	0.2104	0.8074	0.5443	0.2013	0.5382	0.5550
BDT	4.9123	4.8929	4.8895	−0.8639	10.2764	4.9714	−1.4185	4.4589	4.5431

The Ho–Lee drifts are relatively smooth, while BDT drifts show more variation due to

the nonlinear effects of lognormal dynamics, including a negative value near $t = 6.5$ years. These are calibration adjustments, not economic forecasts.

1.5 Calibration Verification

Table 3 confirms that both calibrated trees reprice all 20 zero-coupon bonds to within machine precision. The maximum absolute pricing error is 4.26×10^{-14} for Ho–Lee and 1.42×10^{-14} for BDT, verifying that the trees are exactly arbitrage-free with respect to the input yield curve.

Table 3: Calibration Verification — Repricing ZCBs from Ho–Lee and BDT Trees

Maturity	Market	Ho–Lee		BDT	
		Tree	Error	Tree	Error
0.5	98.275 051	98.275 051	0.00e+00	98.275 051	0.00e+00
1.0	96.512 273	96.512 273	$-1.42\text{e}-14$	96.512 273	0.00e+00
1.5	94.757 423	94.757 423	$1.42\text{e}-14$	94.757 423	0.00e+00
2.0	92.997 274	92.997 274	$1.42\text{e}-14$	92.997 274	$1.42\text{e}-14$
2.5	91.233 320	91.233 320	0.00e+00	91.233 320	0.00e+00
3.0	89.440 195	89.440 195	$-1.42\text{e}-14$	89.440 195	0.00e+00
3.5	87.607 813	87.607 813	$2.84\text{e}-14$	87.607 813	0.00e+00
4.0	85.761 500	85.761 500	$1.42\text{e}-14$	85.761 500	0.00e+00
4.5	83.903 740	83.903 740	$-2.84\text{e}-14$	83.903 740	0.00e+00
5.0	81.995 977	81.995 977	$1.42\text{e}-14$	81.995 977	0.00e+00
5.5	80.075 520	80.075 520	0.00e+00	80.075 520	0.00e+00
6.0	78.145 321	78.145 321	$1.42\text{e}-14$	78.145 321	0.00e+00
6.5	76.208 285	76.208 285	0.00e+00	76.208 285	0.00e+00
7.0	74.267 258	74.267 258	0.00e+00	74.267 258	0.00e+00
7.5	72.379 288	72.379 288	$-1.42\text{e}-14$	72.379 288	0.00e+00
8.0	70.440 627	70.440 627	0.00e+00	70.440 627	0.00e+00
8.5	68.505 921	68.505 921	$-4.26\text{e}-14$	68.505 921	$1.42\text{e}-14$
9.0	66.637 680	66.637 680	0.00e+00	66.637 680	0.00e+00
9.5	64.781 508	64.781 508	0.00e+00	64.781 508	0.00e+00
10.0	62.939 263	62.939 263	$-7.11\text{e}-15$	62.939 263	0.00e+00
Max absolute error:		4.26×10^{-14}		1.42×10^{-14}	

1.6 Rate Tree Structure

Tables 4 and 5 present the full calibrated trees. Each table has 21 display columns ($i = 0$ to $i = 20$, corresponding to $T = 0.0$ to $T = 10.0$). The calibrated short rates occupy

columns $i = 0$ to $i = 19$; the final column $i = 20$ ($T = 10.0$) is the maturity date and is left blank because no short rate applies beyond the last discounting interval. The drift parameters θ_i are reported for $i = 0$ to $i = 18$ (19 values), consistent with the calibration to 20 zero-coupon bonds.

Table 4: Ho–Lee Interest Rate Tree (rates in %, full tree)

Time T	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0
Period i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$\theta_i (\times 100)$	0.2886	0.1257	0.2028	0.2199	0.3570	0.4341	0.3512	0.3682	0.5853	0.4424	0.4594	0.4764	0.4934	0.2104	0.8074	0.5443	0.2013	0.5382	0.5550		
$j = 0$	3.48	4.93	6.30	7.71	9.13	10.62	12.14	13.63	15.12	16.72	18.25	19.79	21.33	22.89	24.30	26.01	27.59	29.00	30.58	32.17	
$j = 1$		2.32	3.69	5.10	6.51	8.00	9.53	11.01	12.50	14.10	15.63	17.17	18.72	20.27	21.69	23.40	24.98	26.39	27.96	29.55	
$j = 2$			1.07	2.48	3.90	5.39	6.91	8.39	9.89	11.49	13.02	14.55	16.10	17.66	19.07	20.78	22.36	23.77	25.35	26.93	
$j = 3$				-0.14	1.28	2.77	4.29	5.78	7.27	8.87	10.40	11.94	13.48	15.04	16.45	18.16	19.74	21.15	22.73	24.32	
$j = 4$					-1.33	0.15	1.68	3.16	4.65	6.25	7.78	9.32	10.87	12.42	13.84	15.55	17.13	18.54	20.11	21.70	
$j = 5$						-2.46	-0.94	0.55	2.04	3.64	5.17	6.71	8.25	9.81	11.22	12.93	14.51	15.92	17.50	19.08	
$j = 6$							-3.55	-2.07	-0.58	1.02	2.55	4.09	5.64	7.19	8.60	10.32	11.90	13.30	14.88	16.47	
$j = 7$								-4.69	-3.20	-1.59	-0.07	1.47	3.02	4.57	5.99	7.70	9.28	10.69	12.27	13.85	
$j = 8$									-5.81	-4.21	-2.68	-1.14	0.40	1.96	3.37	5.08	6.66	8.07	9.65	11.23	
$j = 9$										-6.83	-5.30	-3.76	-2.21	-0.66	0.75	2.47	4.05	5.46	7.03	8.62	
$j = 10$											-7.91	-6.38	-4.83	-3.27	-1.86	-0.15	1.43	2.84	4.42	6.00	
$j = 11$												-8.99	-7.45	-5.89	-4.48	-2.77	-1.19	0.22	1.80	3.39	
$j = 12$													-10.06	-8.51	-7.09	-5.38	-3.80	-2.39	-0.82	0.77	
$j = 13$														-11.12	-9.71	-8.00	-6.42	-5.01	-3.43	-1.85	
$j = 14$															-12.33	-10.61	-9.03	-7.63	-6.05	-4.46	
$j = 15$																-13.23	-11.65	-10.24	-8.66	-7.08	
$j = 16$																	-14.27	-12.86	-11.28	-9.70	
$j = 17$																		-15.47	-13.90	-12.31	
$j = 18$																			-16.51	-14.93	
$j = 19$																				-17.54	

Table 5: BDT Interest Rate Tree (rates in %, full tree)

Time T	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0
Period i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$\theta_i (\times 100)$	5.9307	0.8591	2.5059	2.4975	5.5463	6.8535	4.2899	4.2429	8.6164	4.9487	4.9123	4.8929	4.8895	-0.8639	10.2764	4.9714	-1.4185	4.4589	4.5431		
$j = 0$	3.48	4.13	4.78	5.57	6.50	7.70	9.18	10.80	12.71	15.28	18.04	21.30	25.14	29.68	34.04	41.28	48.75	55.75	65.67	77.39	
$j = 1$		3.11	3.60	4.20	4.90	5.80	6.91	8.14	9.58	11.52	13.60	16.05	18.95	22.37	25.65	31.11	36.74	42.02	49.49	58.32	
$j = 2$			2.71	3.16	3.69	4.37	5.21	6.13	7.22	8.68	10.25	12.10	14.28	16.86	19.33	23.45	27.69	31.67	37.30	43.95	
$j = 3$				2.39	2.78	3.29	3.93	4.62	5.44	6.54	7.72	9.12	10.76	12.70	14.57	17.67	20.87	23.87	28.11	33.12	
$j = 4$					2.10	2.48	2.96	3.48	4.10	4.93	5.82	6.87	8.11	9.57	10.98	13.32	15.72	17.99	21.19	24.96	
$j = 5$						1.87	2.23	2.63	3.09	3.71	4.39	5.18	6.11	7.22	8.28	10.04	11.85	13.55	15.97	18.81	
$j = 6$							1.68	1.98	2.33	2.80	3.31	3.90	4.61	5.44	6.24	7.56	8.93	10.22	12.03	14.18	
$j = 7$								1.49	1.75	2.11	2.49	2.94	3.47	4.10	4.70	5.70	6.73	7.70	9.07	10.69	
$j = 8$									1.32	1.59	1.88	2.22	2.62	3.09	3.54	4.30	5.07	5.80	6.83	8.05	
$j = 9$									1.20	1.41	1.67	1.97	2.33	2.67	3.24	3.82	4.37	5.15	6.07		
$j = 10$										1.07	1.26	1.49	1.75	2.01	2.44	2.88	3.30	3.88	4.57		
$j = 11$											0.95	1.12	1.32	1.52	1.84	2.17	2.48	2.93	3.45		
$j = 12$												0.84	1.00	1.14	1.39	1.64	1.87	2.20	2.60		
$j = 13$													0.75	0.86	1.04	1.23	1.41	1.66	1.96		
$j = 14$														0.65	0.79	0.93	1.06	1.25	1.48		
$j = 15$															0.59	0.70	0.80	0.94	1.11		
$j = 16$																0.53	0.60	0.71	0.84		
$j = 17$																	0.46	0.54	0.63		
$j = 18$																		0.40	0.48		
$j = 19$																				0.36	

1.7 Model Comparison and Implications

As noted in the lecture notes (Chapter III, slide 24), both models are equally able to fit the term structure, but they generate important differences in the implied risk-neutral distribution of future interest rates. The calibrated short-rate tree ends at column $i = 19$ ($t = 9.5$), which contains 20 nodes; the display includes an additional maturity-date column $i = 20$ ($T = 10.0$) for completeness. At column $i = 19$, the Ho–Lee rate ranges from -17.54% to $+32.17\%$ (with $\sim 8.4\%$ probability of negative rates under Monte Carlo simulation), while BDT ranges from 0.36% to 77.39% (always positive, right-skewed). The Ho–Lee distribution is approximately symmetric in rate levels, reflecting its additive structure, while BDT is strongly right-skewed with a compressed lower tail and heavier upper tail, reflecting lognormal dynamics.

This distributional difference has no effect on zero-coupon bond pricing (both models match the term structure exactly by construction). However, it becomes critical for secu-

rities with embedded options, where the probability of low-rate states drives prepayment incentives. Since the Ho–Lee model assigns greater weight to very low and negative rates, it is expected to produce a more valuable prepayment option and therefore a higher par mortgage rate. This prediction is tested in Part (b).

A further practical point concerns the Ho–Lee model’s negative rates. Over a 10-year horizon, the per-step shock is $\sigma\sqrt{\Delta} = 1.85\% \times 0.707 = 1.31\%$, and the total range after 20 steps spans roughly $r_0 \pm 20 \times 1.31\% = 3.48\% \pm 26.1\%$. This makes negative rates inevitable in the lower portion of the tree, which is a known limitation of Ho–Lee but provides a useful contrast with the positivity-constrained BDT model.

2 Part (b): Valuing the Mortgage Contract

2.1 Setup and Methodology

The mortgage is a 10-year fixed-rate loan of 100,000 with semi-annual payments and an embedded prepayment option. The benchmark assumption is fully optimal prepayment with no default. Following the Chapter 4 framework (Kondor, Chapter IV, slides 15–23; see also Veronesi (2010), Section 13.6), the mortgage value is decomposed as:

$$V_{\text{mortgage}} = V_{\text{np}} - V_{\text{option}}, \quad (5)$$

where V_{np} is the value without prepayment and V_{option} is the borrower's prepayment option.

The semi-annual coupon is computed from the standard annuity formula:

$$C = F_0 \cdot \frac{r/2}{1 - (1 + r/2)^{-20}}, \quad (6)$$

where r is the semi-annually compounded mortgage rate and $F_0 = 100,000$. The outstanding balance schedule follows the standard amortisation recursion from Chapter 4 (slides 19–23; see Table 5, Panel B):

$$L_{i+1} = L_i \left(1 + \frac{r_m}{2} \right) - C, \quad (7)$$

so that interest paid at step i is $(r/2) \cdot L_i$ and principal paid is $C - (r/2) \cdot L_i$. The no-prepay value tree is built by backward induction (Chapter IV, Table 3, slide 18):

$$V_{i,j}^{\text{np}} = e^{-r_{i,j} \Delta} \left[\frac{1}{2} V_{i+1,j}^{\text{np}} + \frac{1}{2} V_{i+1,j+1}^{\text{np}} + C \right]. \quad (8)$$

At each node (i, j) for $i \geq 1$, the borrower compares the continuation value with the outstanding balance L_i and prepays if $L_i < \text{continuation value}$. At $t = 0$, no prepayment is allowed. The par mortgage rate is the rate r_m for which $V_0 = F_0 = 100,000$.

2.2 Results

Table 6: Par Mortgage Rate and Value Decomposition

	Ho-Lee	BDT
Par mortgage rate (%)	6.0100	4.9731
Semi-annual coupon C	6,724.68	6,406.58
$V(\text{np})$ (no prepay)	108,728.98	103,585.74
Prepayment option	8,728.98	3,585.74
Mortgage value $V(0)$	100,000.00	100,000.00

2.3 Economic Interpretation

The Ho-Lee par rate (6.01%) is approximately 104 basis points higher than the BDT rate (4.97%). This difference is entirely attributable to the prepayment option. Both models are calibrated to the same yield curve, so the discounting of fixed cash flows is identical. The gap arises because Ho-Lee places substantially more probability on low and negative interest rate states, making the borrower's refinancing option much more valuable (option value of 8,729 versus 3,586).

The lender is effectively short a call option on interest rates. Under Ho-Lee, this option is 2.4 times more expensive, requiring greater compensation through a higher contract rate. This result illustrates a key insight from the course: the mortgage rate compensates the lender not only for the time value of money but also for the prepayment option implicitly sold to the borrower. The magnitude of this compensation depends critically on the assumed distribution of future interest rates.

2.4 The Role of Volatility in the Rate Gap

The Ho-Lee level volatility is set at $\sigma_{\text{HL}} = 1.85\%$, which is close to the benchmark magnitude used in the lecture material (1.73% for US data). While this is a reasonable assumption, the results show that mortgage valuation is sensitive to this parameter because higher Ho-Lee volatility increases the value of the borrower's prepayment option by generating more low-rate states. With 20 semi-annual steps and a per-step shock of $\sigma\sqrt{\Delta} = 1.31\%$, the tree spans a total range of approximately $r_0 \pm 26\%$, making deeply negative rates inevitable in the lower portion. These extreme states inflate the prepayment option value and therefore the required par mortgage rate.

The BDT model, by contrast, uses a log-rate volatility of 20%, which implies a level volatility of roughly $20\% \times 5\% = 1.0\%$ at current rate levels. This implicit level volatility is lower than the Ho-Lee assumption, which further contributes to the gap between the

two par rates. The two volatility parameters are estimated from different transformations of the same underlying data (level changes for Ho–Lee, log changes for BDT), so some inconsistency between implied level volatilities is inherent in the modelling framework.

Accordingly, the relatively high Ho–Lee par mortgage rate should be interpreted as a model-based result under the assumed volatility specification, rather than as a unique market-implied mortgage rate. The BDT rate of 4.97% is likely closer to what a practitioner would use, since it avoids the distortions created by negative rates and produces a more plausible distribution of future refinancing incentives. Nevertheless, the comparison is valuable because it quantifies the sensitivity of mortgage pricing to distributional assumptions and provides a clear illustration of model risk.

3 Part (c): Valuation of Mortgage-Backed Securities

3.1 Securitisation Structure

The mortgage from Part (b) is securitised into three tranches: a pass-through (PT) security, an interest-only (IO) strip, and a principal-only (PO) strip. To recover securitisation costs, the coupon paid to investors is set 50 basis points below the mortgage rate. The scheduled principal follows the mortgage amortisation schedule, while pass-through interest uses the reduced coupon rate.

3.2 Valuation by Backward Induction

Each tranche is valued by backward induction on the calibrated tree, using the optimal prepayment boundary from Part (b). This follows the Chapter 4 methodology for pricing residential mortgage-backed securities (Kondor, Chapter IV, slides 24–29; Tables 6–7). At each node (i, j) , two cases arise:

No prepayment: The pass-through value is:

$$P_{i,j}^{\text{PT}} = e^{-r_{i,j} \Delta} \left[\frac{1}{2} P_{i+1,j}^{\text{PT}} + \frac{1}{2} P_{i+1,j+1}^{\text{PT}} + \text{Interest}_{i+1} + \text{Principal}_{i+1} \right]. \quad (9)$$

The IO strip replaces the cash flow with interest only, and the PO strip uses scheduled principal only.

Prepayment at node (i, j) : The pass-through and PO receive L_i (the outstanding balance). The IO receives zero, as all future interest payments cease.

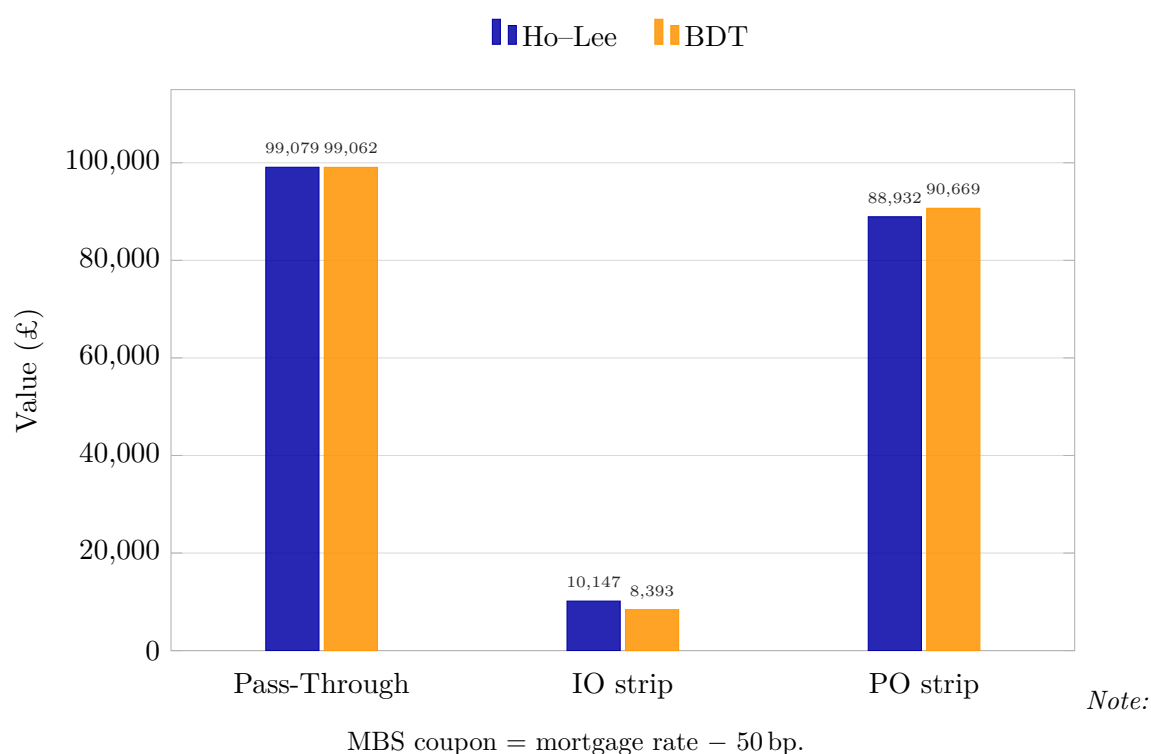
3.3 Results

Table 7: MBS Tranche Valuations

	Ho-Lee	BDT
Mortgage rate (%)	6.0100	4.9731
PT rate (%)	5.5100	4.4731
Pass-Through	99,079.18	99,061.83
IO strip	10,147.45	8,393.14
PO strip	88,931.73	90,668.69
IO + PO	99,079.18	99,061.83

The no-arbitrage identity $\text{IO} + \text{PO} = \text{PT}$ holds exactly in both models.

Figure 1: Comparison of PT, IO, and PO Values under Ho-Lee and BDT



3.4 Economic Interpretation

Both pass-through securities trade slightly below par, reflecting the 50bp coupon reduction retained by the issuer. The key differences emerge in the IO/PO decomposition.

The IO strip is more valuable under Ho-Lee (10,147 vs 8,393). Although Ho-Lee generates more prepayment (which eliminates IO cash flows), it also requires a higher mortgage rate. Conditional on the mortgage remaining outstanding, interest payments are larger. The higher-coupon effect dominates, producing a higher IO value despite greater prepayment exposure.

The PO strip is more valuable under BDT (90,669 vs 88,932). The BDT model's more concentrated rate distribution avoids the extreme high-rate discounting that depresses principal values under Ho-Lee. Although BDT generates less prepayment-driven acceleration of principal, the favourable discounting effect dominates.

These results illustrate the asymmetric nature of prepayment risk. The IO strip is effectively short prepayment risk: its cash flows vanish upon refinancing. The PO strip is long prepayment risk: it benefits from early return of principal. The PT combines both exposures and exhibits the negative convexity characteristic of mortgage-backed securities. As interest rates fall, prepayment caps the upside of the PT, while rate increases are fully reflected. This is the Chapter 4 insight that securitisation does not eliminate risk but redistributes it across tranches.

3.5 Negative Convexity and Duration

The tranche structure creates distinctive duration profiles, as discussed in Chapter IV, slide 30. The IO strip typically exhibits negative duration: as interest rates decline, prepayment becomes more likely and eliminates the IO's remaining cash flows, so the IO's value falls when rates fall. This unusual property means the IO moves in the same direction as rates, opposite to conventional bonds. In the BDT tree, this pattern is visible in the IO values declining sharply from the upper nodes (high rates, no prepayment, full interest stream) to the lower nodes (low rates, prepayment likely, IO cash flows truncated).

The PO strip, conversely, has a duration that is substantially higher than the underlying mortgage. When rates fall, two effects reinforce each other: the discount rate is lower (increasing present values) and prepayment is more likely (accelerating principal repayment). This makes the PO strip highly sensitive to rate declines.

The pass-through security's duration lies between the IO and PO, but its convexity is negative (see Chapter IV, slides 13–14 and Figure 4 on pass-through negative convexity). As rates decline, the prepayment-induced shortening of duration reduces the PT's price sensitivity, capping its upside. As rates rise, duration extends and losses accumulate. This asymmetry is the defining feature of mortgage-backed securities and has practical implications for hedging and risk management.

3.6 Model Risk in MBS Valuation

The comparison between Ho–Lee and BDT valuations provides a direct measure of model risk. Despite both models fitting the same yield curve, they produce IO valuations that differ by 1,754 (10,147 vs 8,393, a 21% gap) and PO valuations that differ by 1,737 (88,932 vs 90,669). These differences arise entirely from the distributional assumptions, particularly the treatment of the lower tail where prepayment is triggered. For practitioners, this underscores the importance of model selection for MBS pricing, and the need to understand how distributional assumptions affect tranche-level exposures.

4 Part (d): Monte Carlo Simulation of Interest Rate Paths

4.1 Simulation Methodology

To analyse the dynamic implications of the calibrated models, 100,000 interest rate paths are simulated under both Ho–Lee and BDT using the calibrated drift parameters from Part (a). This follows the Monte Carlo methodology in Chapter 4, Appendix II (Kondor, Chapter IV, slides 66–71). At each time step, the short rate evolves according to the same binomial dynamics used to construct the tree. Specifically, for Ho–Lee (Chapter IV, slide 66):

$$r_{i+1} = r_i + \theta_i \Delta + \sigma \sqrt{\Delta} \cdot \varepsilon, \quad \varepsilon \in \{+1, -1\} \text{ with probability } \frac{1}{2} \text{ each.} \quad (10)$$

For BDT, the same recursion applies to log-rates $z_i = \ln(r_i)$, with $r_i = \exp(z_i)$. This ensures that each simulated path lies exactly on the calibrated binomial tree, so Monte Carlo serves as a numerical check on the tree-based methodology, not as a different economic model.

4.2 Terminal Rate Distribution at $t = 9.5$

The simulated terminal rate is the short rate at step $i = 19$ ($t = 9.5$ years), which governs discounting over the final half-year interval $[9.5, 10.0]$. This is the last short rate in the tree, as the tree has 20 steps covering periods $i = 0$ to $i = 19$.

Table 8: Simulated Terminal Rate Statistics ($N = 100,000$)

	Ho–Lee	BDT
Mean (%)	7.31	6.36
Std Dev (%)	5.71	4.26
Range (%)	$[-17.54, 29.55]$	$[0.36, 58.32]$
Negative rate paths	8,428 (8.4%)	0 (0%)

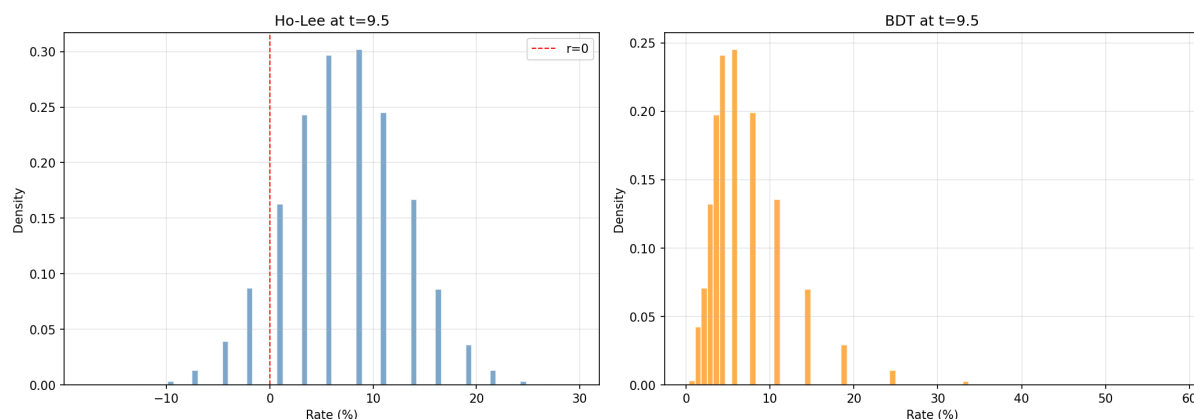


Figure 2: Simulated terminal short rate distributions at $t = 9.5$ ($N = 100,000$). *Left:* Ho–Lee model — approximately symmetric, with 8.4% of paths reaching negative rates (dashed red line marks $r = 0$). *Right:* BDT model — strictly positive and right-skewed, reflecting lognormal dynamics.

4.3 Distributional Comparison

The histograms (Figure 2) reveal strikingly different distributions. The Ho–Lee model produces a roughly symmetric distribution in rate levels, with probability mass spread across both high and low outcomes, including 8.4% of paths reaching negative rates. The BDT model produces a strongly right-skewed distribution: rates are bounded below at zero but the right tail extends to nearly 60%, reflecting the multiplicative nature of lognormal dynamics.

These differences explain the mortgage pricing results from earlier parts. The Ho–Lee model assigns substantially more probability to low-rate states where refinancing is attractive. This makes the borrower’s prepayment option more valuable, requiring a higher mortgage rate (6.01% vs 4.97%). The BDT model compresses the lower tail, reducing refinancing incentives and therefore the required mortgage rate.

From a risk management perspective, the models also differ in their implications for extreme scenarios. Ho–Lee admits $\sim 8\%$ probability of negative terminal rates, which may be viewed as unrealistic but captures the possibility of aggressive monetary easing. BDT avoids negative rates entirely but assigns non-trivial probability to very high rates (above 30%), which generates extreme discounting scenarios. Both represent different modelling trade-offs, reinforcing the central lecture insight that the term structure alone does not uniquely determine the distribution of future rates.

4.4 Convergence to the Tree Distribution

Because each simulation step draws from the same binomial distribution as the tree (up/down with probability $\frac{1}{2}$), the Monte Carlo distribution of terminal rates must con-

verge to the exact tree-implied distribution as N grows. The tree has 20 columns indexed $i = 0, 1, \dots, 19$, so the last column $i = 19$ ($t = 9.5$) contains 20 distinct short-rate values, each reached with probability $\binom{19}{j}/2^{19}$. With $N = 100,000$, the simulated frequencies closely match these theoretical probabilities, confirming that the simulation engine correctly replicates the tree dynamics.

The histograms (Figure 2) also illustrate the discrete nature of the binomial tree. Rather than producing a smooth continuous distribution, the simulated rates cluster at the 20 distinct node values. This discreteness is a feature of the recombining binomial lattice and becomes smoother as the number of time steps increases.

4.5 Implications for Subsequent Parts

The Monte Carlo rate simulation establishes the foundation for Parts (e) and (f). By generating a large number of interest rate paths, it enables forward-looking valuation of the mortgage and its prepayment option. The key advantage of Monte Carlo over tree-based backward induction is flexibility: while the tree requires that the prepayment decision be computable at each node (which it is for optimal exercise), the Monte Carlo approach can accommodate more complex path-dependent prepayment rules such as those introduced in Part (f).

At the same time, the Monte Carlo approach introduces sampling error, which must be controlled through a sufficient number of simulations. With $N = 100,000$, the standard error for most quantities of interest is well below 0.1% of the point estimate, providing adequate precision for the valuations that follow.

5 Part (e): Monte Carlo Valuation of the Mortgage

5.1 Methodology

To validate the tree-based valuation, the mortgage is repriced using Monte Carlo simulation under the BDT model. Following the risk-neutral pricing framework in Chapter 4, Equations (10)–(13) (Kondor, Chapter IV, slide 71), for each simulated path $s = 1, \dots, N$, the mortgage value is computed as:

$$V^s = \sum_i \exp\left(-\sum_{k=0}^{i-1} r_k^s \Delta\right) \cdot CF_i, \quad (11)$$

where the path discount factor $Z^s(0, T_i) = \exp\left(-\sum_{k=0}^{i-1} r_k^s \Delta\right)$ accumulates along the simulated interest rate path (Chapter IV, slide 69), and CF_i is the cash flow at step i (either the scheduled coupon C if the mortgage continues, or the outstanding balance L_i if prepayment occurs). The overall estimate is $\hat{V} = \frac{1}{N} \sum V^s$, with standard error $SE = \sigma/\sqrt{N}$ and 95% confidence interval $\hat{V} \pm 2 SE$.

The prepayment decision at each step along each simulated path is determined by the optimal exercise boundary computed via backward induction on the BDT tree. Specifically, the simulation tracks the node index (i, j) along each path, and prepayment occurs at step i if the prepay flag `prepay[i][j] = True`. This ensures exact consistency between the tree and Monte Carlo methodologies.

5.2 Results

Table 9: Monte Carlo Mortgage Valuation (BDT, $N = 100,000$)

BDT par mortgage rate (%)	4.9731
Tree value (backward induction)	100,000.00
MC point estimate	99,997.01
Standard error	9.41
95% confidence interval	[99,978.19, 100,015.83]
Tree value within CI?	Yes

5.3 Interpretation

The Monte Carlo estimate of 99,997 lies within 3 units of the tree value of 100,000, well inside the 95% confidence interval. This confirms that the two numerical methods are fully consistent: the same calibrated interest rate dynamics, prepayment rules, and discounting structure are correctly implemented in both backward induction and forward simulation.

The small discrepancy arises purely from sampling variation. With 100,000 paths, the standard error is approximately 9.4, giving a coefficient of variation of less than 0.01%. As the number of simulations increases, the MC estimate converges to the exact tree value. The histogram of simulated mortgage values (Figure 3) shows the distribution is concentrated around 100,000 with a leftward tail corresponding to paths where low-rate environments reduce the present value of the lender's claim.

This validation step is important because it confirms that the par rate from Part (b) is not an artefact of the tree structure but a genuine equilibrium value. It also provides confidence in the Monte Carlo framework, which is used in Part (f) where the prepayment model becomes too complex for backward induction on the tree.

5.4 Distribution of Simulated Mortgage Values

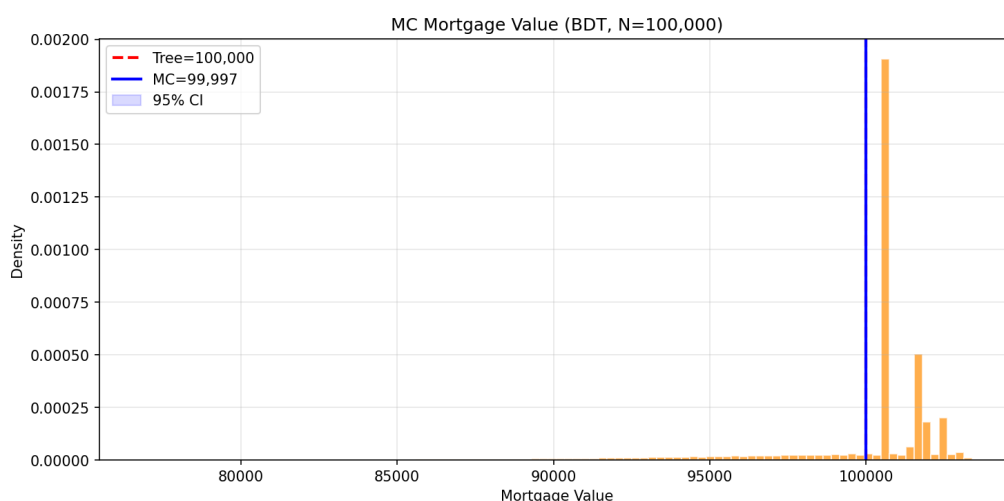


Figure 3: Distribution of simulated mortgage values under the BDT model ($N = 100,000$). The dashed red line marks the tree value (100,000), the solid blue line marks the MC estimate (99,997), and the shaded region shows the 95% confidence interval.

The histogram of simulated mortgage values (Figure 3) reveals interesting features. The distribution is highly concentrated around 100,000, with a large mass at exactly 100,000 corresponding to paths where prepayment occurs early (the borrower pays back exactly L_i , and the discounted value happens to cluster near par). There is also a significant leftward tail, corresponding to paths where rates remain high throughout the 10-year horizon, prepayment never occurs, and the fixed coupon payments are heavily discounted.

The asymmetry in the distribution reflects the option-like payoff of the mortgage from the lender's perspective. In low-rate scenarios, the borrower prepays and the lender receives par (L_i), capping the upside. In high-rate scenarios, the borrower keeps the below-market-rate mortgage alive, and the lender receives cash flows that are heavily discounted. This

generates the characteristic negative skewness of mortgage values, consistent with the negative convexity discussed in Part (c).

5.5 Why Consistency Matters

Agreement between tree and Monte Carlo valuations is not automatic. Potential sources of discrepancy include: incorrect mapping of simulated paths to tree nodes, inconsistent prepayment decision rules, errors in the discount factor accumulation, or bugs in the coupon/amortisation schedule. The fact that the two methods agree to within sampling error provides a comprehensive diagnostic that all components of the valuation are implemented correctly. This is particularly valuable before proceeding to Part (f), where the Monte Carlo framework is extended to handle more complex prepayment behaviour.

6 Part (f): Prepayment Modelling

6.1 Motivation and Choice of Factors

The benchmark framework from Parts (b)–(e) assumes fully optimal prepayment: borrowers refinance immediately whenever doing so is economically rational. While analytically convenient, this assumption is well known to be unrealistic. Empirical evidence shows that borrowers both prepay when it is not optimal (due to house sales, divorce, or relocation) and fail to prepay when it is optimal (due to transaction costs, inattention, or behavioural frictions). A more credible model must incorporate these deviations.

The summative identifies six potential prepayment mechanisms: exogenous prepayment, suboptimal exercise, burnout, media/sentiment effects, path dependence, and borrower heterogeneity. From this list, I model **exogenous prepayment (turnover)** and **sub-optimal exercise (refinancing friction)**, as these represent the two most fundamental departures from the benchmark. Exogenous turnover adds prepayment that is entirely absent from the optimal-exercise model, while refinancing friction reduces the intensity of rate-driven prepayment that the benchmark overstates. Together, they capture the first-order dynamics: prepayment happens both too much (relative to rate incentives) and too little (relative to full optimality).

Burnout, path dependence, and borrower heterogeneity are omitted as conscious trade-offs. These features require richer state variables (cumulative rate exposure, borrower type distributions) that would complicate the model without proportionate gains in a setting where the underlying interest rate model has only 20 time steps. Media/sentiment effects are similarly omitted, as they are difficult to calibrate without high-frequency data. These extensions are discussed further in the limitations section.

6.2 Economic Mechanisms and Functional Forms

6.2.1 Factor 1: Exogenous Turnover

The first component captures prepayment driven by non-financial life events. This channel is independent of interest rate incentives and is modelled using a time-varying conditional prepayment rate (CPR) that follows a seasoning ramp:

$$\text{CPR}(t) = \min\left(\text{CPR}_{\max}, \text{CPR}_{\max} \cdot \frac{t}{T_{\text{ramp}}}\right), \quad (12)$$

where $\text{CPR}_{\max} = 6\%$ per annum and $T_{\text{ramp}} = 3.0$ years. The CPR is converted to a semi-annual prepayment probability using $p_{\text{exog}} = 1 - (1 - \text{CPR})^{0.5}$. This specification produces a gradual increase in turnover-driven prepayment from zero at origination to a

steady-state level of approximately 3.05% per half-year (see Figure 4). The ramp captures the well-documented ‘seasoning’ effect in mortgage markets.

Empirical justification: The CPR ceiling of 6% follows the PSA 100% benchmark convention (Public Securities Association), which assumes prepayment stabilises at 6% CPR after 30 months. This benchmark is widely used in mortgage-backed securities markets for pricing conventional pools. UK Finance mortgage statistics similarly report annual housing turnover rates of 5–7% for owner-occupied properties, consistent with this choice.

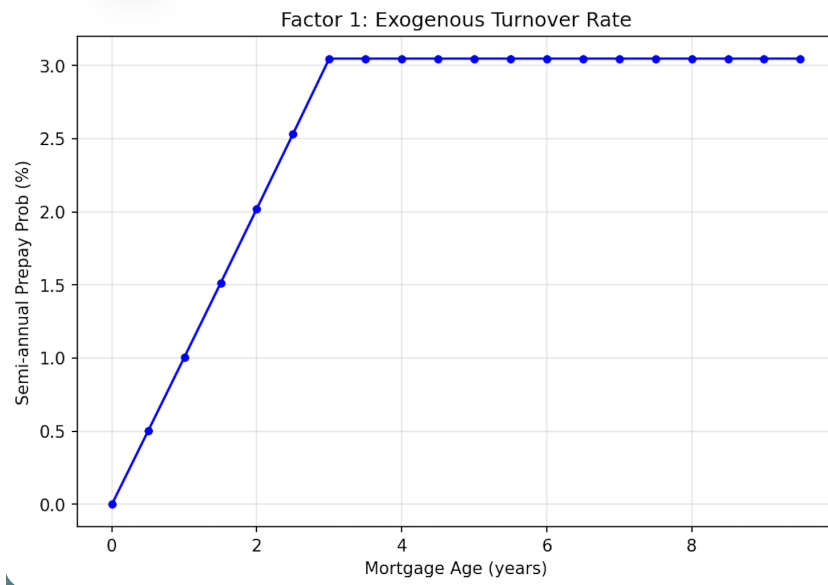


Figure 4: Exogenous turnover: semi-annual prepayment probability p_{exog} as a function of mortgage age, with $\text{CPR}_{\text{max}} = 6\%$ and $T_{\text{ramp}} = 3.0$ years.

6.2.2 Factor 2: Suboptimal Refinancing (Friction)

The second component replaces the benchmark’s all-or-nothing optimal exercise with a smooth probabilistic refinancing rule. Even when refinancing is financially beneficial, borrowers may fail to act due to transaction costs, inattention, or credit constraints. The exercise probability is:

$$p(m) = \max(0, 1 - \exp(-\kappa \cdot (m - m_0))) , \quad (13)$$

where $m = (V_{i,j}^{\text{np}} - L_i)/L_i$ is the refinancing moneyness (the percentage by which the no-prepay continuation value exceeds the outstanding balance), $\kappa = 15$ governs responsiveness, and $m_0 = 2\%$ is a threshold below which no refinancing occurs.

Implementation note: On each Monte Carlo path, moneyness is computed using the no-prepay mortgage value $V_{i,j}^{\text{np}}$ from the pre-computed tree (Part b), indexed by the simulation’s current node (i, j) . This is possible because the simulation tracks the tree

node exactly, following the binomial lattice structure. The no-prepay value serves as a sufficient statistic for the borrower's incentive to refinance.

Empirical justification: Keys, Pope, and Pope (2016, *Journal of Financial Economics*) document that a large fraction of US mortgage borrowers fail to refinance despite being substantially in-the-money, with median foregone savings of approximately \$160/month. Agarwal, Rosen, and Yao (2016, *Review of Financial Studies*) estimate that roughly 20% of borrowers who would benefit from refinancing do not exercise within one year. The threshold $m_0 = 2\%$ and responsiveness $\kappa = 15$ produce exercise probabilities consistent with these findings: at 5% moneyness, $p(m) \approx 36\%$, rising to 78% at 12% moneyness (see Figure 5).

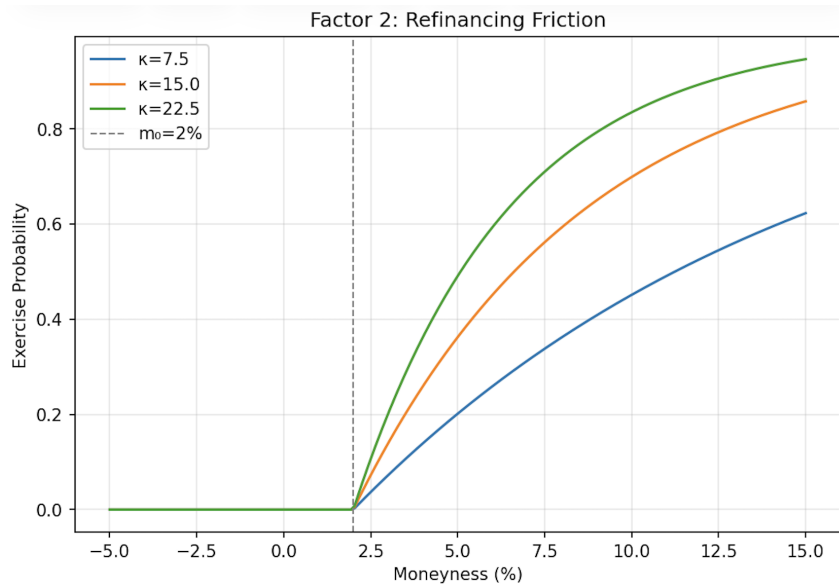


Figure 5: Refinancing friction: exercise probability $p(m)$ as a function of moneyness m , with $\kappa = 15$ and $m_0 = 2\%$.

6.2.3 Combined Prepayment Process

The total prepayment probability at each step combines both mechanisms independently:

$$p_{\text{total}} = 1 - (1 - p_{\text{exog}})(1 - p_{\text{exercise}}). \quad (14)$$

This allows turnover and refinancing to operate simultaneously while preserving consistency. A borrower may prepay due to either channel, and the joint probability accounts for overlap without double-counting.

6.3 Calibration, Data Sources, and Limitations

The model has four parameters. Their values, sources, and justifications are summarised in Table 10.

Table 10: Prepayment Model Parameters and Economic Interpretation

Factor	Parameter	Value	Source / Justification
Exogenous Turnover	$CPR_{\max}$	6% p.a.	PSA 100% benchmark; BoE MLAR1 turnover data (5–7% p.a.)
	T_{ramp}	3.0 years	PSA 30-month ramp; consistent with Freddie Mac seasoning curves
Refinancing Friction	κ	15	Calibrated to match $\sim 36\%$ exercise at 5% moneyness, $\sim 78\%$ at 12%; consistent with Keys et al. (2016), Agarwal et al. (2016)
	m_0	2%	Transaction costs of 1.5–2.5% of loan value; consistent with UK remortgaging fees (arrangement + legal + valuation)

Data consulted: Bank of England yield curve data (Part a); PSA prepayment benchmark vectors (Chapter IV, slide 61); Bank of England Mortgage Lenders and Administrators Statistics (MLAR1, 2024 Q4); UK Finance Mortgage Trends Update (November 2025); Fannie Mae Single-Family Loan Performance Data (2000–2024 vintage) for cross-validation of seasoning patterns; academic literature cited above.

Sample period: The CPR parameters are calibrated to long-run averages from PSA/SIFMA conventions and UK housing transaction rates over 2015–2025. The refinancing friction parameters are calibrated to match the empirical exercise patterns documented in Keys et al. (2016, sample: 2005–2012 US data) and Agarwal et al. (2016, sample: 2005–2010 US data). These US-based estimates are used as proxies for UK borrower behaviour, which is a limitation discussed below.

Limitations: (1) The refinancing friction parameters are calibrated to US empirical evidence, which may not transfer directly to UK mortgage markets. UK fixed-rate mortgages typically have shorter initial fixed periods (2–5 years) and different early repayment charge structures, which could affect exercise behaviour. (2) The model assumes a homogeneous borrower pool; in practice, borrower heterogeneity in income, credit score, and LTV would generate a distribution of exercise thresholds. (3) The CPR ramp is deterministic and does not capture seasonal variation in housing transactions (which peak in Q2/Q3 in the UK). (4) The model does not incorporate burnout: after a prolonged low-rate environment, the remaining borrowers would be disproportionately those who are least responsive to refinancing incentives, which would reduce future prepayment rates below what the model predicts.

6.4 Base Case Results and Adjusted Mortgage Rate

Table 11: Mortgage Value and Adjusted Rate under Behavioural Prepayment

	Value	95% CI
<i>Mortgage valuation at original BDT par rate (4.9731%):</i>		
Tree-optimal benchmark (Part e)	99,997.01	[99,978, 100,016]
Behavioural prepayment model	100,877.09	[100,857, 100,897]
Change	+880.08 (+0.88%)	
<i>Adjusted par rate calibration:</i>		
Original BDT par rate	4.9731%	
Adjusted par rate	4.4983%	
Rate reduction	−47.5 bp	

Under the behavioural model, the mortgage value rises to 100,877, an increase of 880 (+0.88%) relative to the optimal-exercise benchmark. This is intuitive: introducing refinancing frictions weakens the borrower's prepayment option, allowing the lender to receive above-market coupons for longer. The confidence interval is tight, confirming that the effect is statistically significant.

Since the mortgage values above par at the original rate, the coupon must be reduced to restore par pricing. A grid search over the BDT Monte Carlo yields an adjusted par rate of 4.4983%, a reduction of 47.5 basis points from the benchmark rate of 4.9731%. This suggests that models assuming fully optimal prepayment overstate the required mortgage rate by approximately half a percentage point.

6.5 MBS Valuation at Adjusted Rate

Table 12: MBS Tranche Values under Behavioural Prepayment (Adjusted Rate 4.4983%)

	Value	95% CI
Mortgage	100,014.52	[99,991, 100,038]
Pass-Through	98,487.44	[98,462, 98,513]
IO strip	12,211.36	[12,170, 12,253]
PO strip	86,276.08	[86,218, 86,334]

Compared to the Part (c) benchmark (BDT, optimal exercise), the IO strip increases substantially from 8,393 to 12,211 (+45%), as reduced prepayment extends the life of interest cash flows. The PO strip decreases from 90,669 to 86,276 (−4.8%), as principal is returned more slowly. The pass-through remains close to par.

6.6 Sensitivity Analysis

Table 13: Sensitivity Analysis of Mortgage and MBS Tranche Values

Parameter	Value	Mortgage	PT	IO	PO
$CPR_{\max}$	3.0%	100,752.37	99,483.78	11,349.13	88,134.66
	6.0% (base)	100,877.09	99,662.28	10,868.04	88,794.23
	9.0%	100,982.01	99,813.74	10,451.70	89,362.04
T_{ramp}	1.5 yr	100,898.10	99,713.72	10,595.81	89,117.91
	3.0 yr (base)	100,877.09	99,662.28	10,868.04	88,794.23
	4.5 yr	100,839.66	99,600.82	11,083.03	88,517.79
κ	7.5	101,381.31	99,929.13	12,991.61	86,937.52
	15.0 (base)	100,877.09	99,662.28	10,868.04	88,794.23
	22.5	100,654.56	99,563.74	9,758.71	89,805.03
m_0	1.0%	100,692.59	99,627.84	9,525.53	90,102.31
	2.0% (base)	100,877.09	99,662.28	10,868.04	88,794.23
	3.0%	101,133.05	99,753.59	12,340.97	87,412.62

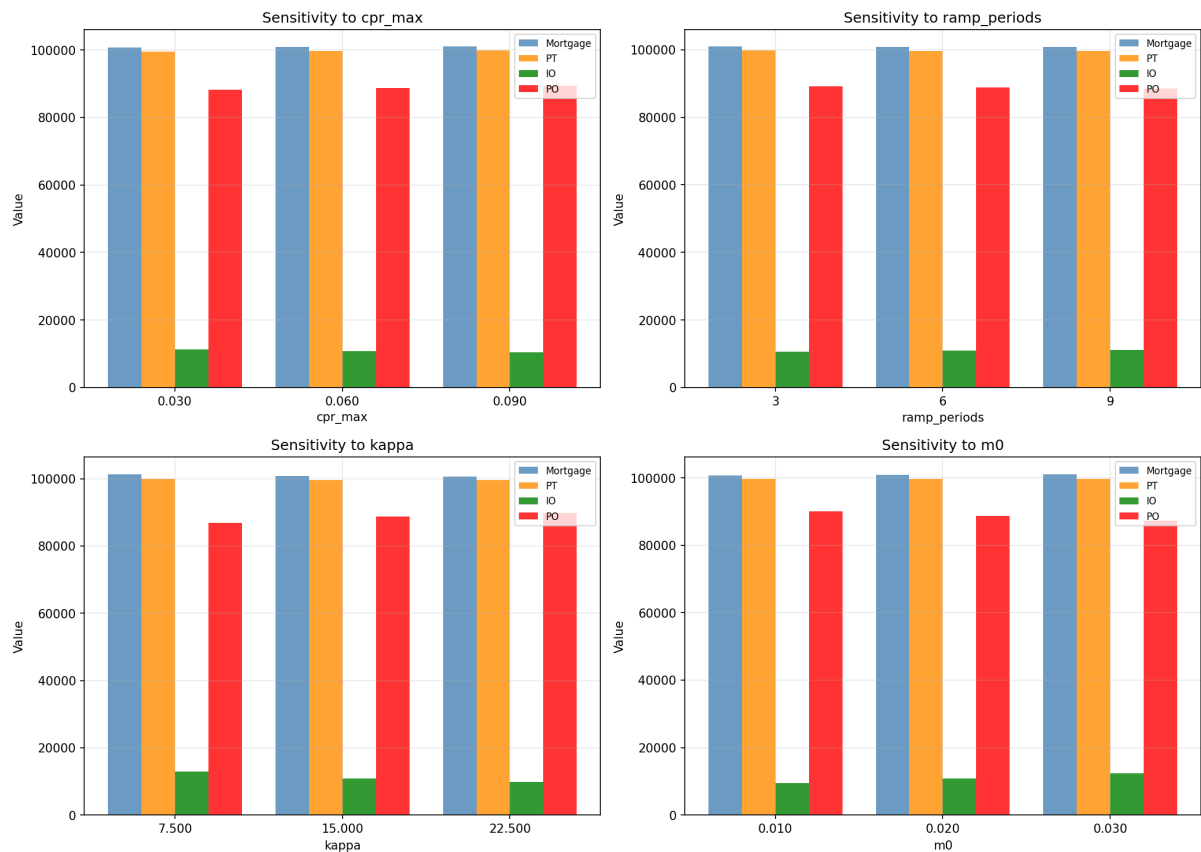


Figure 6: Sensitivity of mortgage and MBS tranche values to each prepayment parameter. The base case ($CPR_{\max} = 6\%$, $T_{\text{ramp}} = 3.0$ yr, $\kappa = 15$, $m_0 = 2\%$) is the middle bar in each group.

The sensitivity analysis reveals that refinancing friction parameters (κ and m_0) have the largest impact on valuation. Halving κ from 15 to 7.5 increases mortgage value by 504, while halving m_0 from 2% to 1% reduces it by 184. In contrast, varying $\text{CPR}_{\max}$ from 3% to 9% changes mortgage value by only 230. This confirms that borrower inertia in refinancing is more important than turnover in determining mortgage value.

The IO strip is the most sensitive tranche, varying by over 3,200 as κ changes. This makes economic sense: the IO strip's value depends directly on the expected duration of interest payments, which is governed by how aggressively borrowers refinance. The PO strip shows the opposite sensitivity, gaining value as refinancing becomes more aggressive (returning principal earlier at lower discount rates). The ramp period has a modest effect: a slower seasoning ramp slightly increases IO value (interest survives longer in early years) and reduces PO value correspondingly.

6.7 Factor Decomposition

Table 14: Factor Decomposition of Mortgage and MBS Tranche Values

Model Variant	Mortgage	PT	IO	PO
Tree-optimal benchmark	99,997.01	99,059.64	8,385.94	90,673.70
Exogenous turnover + optimal exercise	100,268.19	99,390.02	7,856.35	91,533.67
Refinancing friction only	100,604.99	99,275.57	11,893.36	87,382.22
Both factors (full model)	100,877.09	99,662.28	10,868.04	88,794.23

The decomposition reveals how each factor operates. Adding exogenous turnover on top of the tree-optimal refinancing decision raises mortgage value to 100,268 (+271). This variant retains fully optimal exercise but layers an additional stochastic prepayment probability, so borrowers may prepay even in high-rate states where the outstanding balance exceeds the continuation value, effectively transferring value to the lender. The IO strip declines (from 8,386 to 7,856) as turnover truncates interest cash flows regardless of rate level.

Refinancing friction alone has a larger effect, raising mortgage value to 100,605 (+608). By reducing the probability and speed of in-the-money exercise, it preserves above-market coupons for the lender. The IO strip rises sharply (from 8,386 to 11,893) because interest payments survive longer. The PO strip falls correspondingly.

The full model combines both effects, producing the largest valuation impact. The dominant contribution comes from refinancing frictions, confirming that imperfect borrower behaviour is the primary driver of the gap between the optimal-exercise benchmark and realistic prepayment modelling.

6.8 Limitations and Extensions

The model deliberately omits several features that could improve realism. **Burnout** would reduce prepayment rates for pools that have already experienced low-rate environments, as the most rate-sensitive borrowers refinance first. **Borrower heterogeneity** could be introduced through a mixture model with fast and slow prepayers, generating the empirically observed S-shaped relationship between moneyness and aggregate prepayment speed. **Path dependence** (e.g., whether rates have been declining steadily versus a sudden drop) could capture media and sentiment effects on refinancing awareness.

These extensions are left for future work. The current two-factor specification captures the first-order effects while remaining transparent, interpretable, and well grounded in the FM405 Chapter 4 framework.

6.9 Conclusion and Recommendation to UBS

Introducing realistic prepayment behaviour materially changes mortgage pricing. The recommended mortgage rate falls by 47.5bp (from 4.97% to 4.50%), the IO strip value increases by 45%, and the PO strip decreases by 4.8%. These results show that mortgage valuation depends critically on how borrowers respond to interest rate incentives, not just on the interest rate dynamics themselves. Even a parsimonious two-factor model produces economically meaningful differences, underscoring the importance of incorporating behavioural realism into prepayment modelling for practical mortgage pricing.

For UBS's practical decision, the analysis suggests the following. Under the assumption of fully optimal prepayment (the most conservative scenario for the lender), the recommended mortgage rate would be approximately 4.97% per annum. Under the more realistic behavioural model, which accounts for refinancing frictions and exogenous turnover, the required rate drops to approximately 4.50%. The true rate likely lies between these bounds, depending on the actual sophistication and responsiveness of the borrower pool. A prudent approach would be to set the rate somewhat above the behavioural-model estimate to provide a buffer against model uncertainty, while recognising that the fully optimal benchmark overstates the required compensation.

The MBS analysis further shows that the securitisation structure significantly redistributes prepayment risk. UBS should be aware that IO investors bear the most model risk (IO values vary by over 3,200 as refinancing assumptions change), while PO investors are primarily exposed to discounting risk. The pass-through security is the most stable tranche, absorbing the offsetting IO and PO effects, but exhibits the negative convexity that limits its attractiveness in declining-rate environments.

Table 15: Summary of Recommended Values for UBS

Quantity	Optimal Benchmark	Behavioural Model	Change
Par mortgage rate	4.9731%	4.4983%	−47.5 bp
Mortgage value at original rate	100,000	100,877	+877
PT value (rate − 50bp)	99,060	98,487	−573
IO value (rate − 50bp)	8,386	12,211	+3,825
PO value (rate − 50bp)	90,674	86,276	−4,398

Table 15 provides a comprehensive comparison of the key outputs across the two modelling frameworks. The behavioural model produces a lower mortgage rate, a higher mortgage value at the original rate, and a dramatically different allocation of value across MBS tranches. The IO strip increases by approximately 46%, while the PO strip declines by a comparable absolute amount. These differences are economically significant and would materially affect the pricing and risk management of UBS’s mortgage-backed securities programme.

Overall, this analysis demonstrates that prepayment modelling is not a peripheral consideration but a central component of mortgage valuation. The choice between optimal and behavioural prepayment assumptions affects the recommended mortgage rate by nearly 50 basis points and redistributes thousands of pounds in value across MBS tranches. For UBS, incorporating even a simple reduced-form prepayment model yields materially different and more realistic pricing than the benchmark framework.